



Principal Architect

Future Architect

Welcome to your adventure!

You have unlocked the Principal Architect mission.
Print this pack or explore the digital version.

Your secret link to the online experience:

[YOUR SECRET UNLOCK URL GOES HERE]

How to Use Your Pack

■ Printable Version

1. Print this PDF
2. Grab coloured pencils, fine liners or markers
3. Work through each challenge page
4. Complete the drawing spaces
5. Write your name at the top
6. Share your amazing designs!

■ Digital Version

1. Visit your secret link
2. Choose 'Principal Architect'
3. Complete the challenges on screen
4. Use the drawing tools
5. Collect your stars and XP
6. Generate your certificate!

■ Pro Tips:

- Both versions are the same – choose what works for your day
- The online version works on phones, tablets, and computers
- Everything syncs automatically – start on paper, finish on screen
- Print from Page 3 onwards (Pages 1–2 are for screen viewing)
- Questions? Visit thetravellabacademy.co.uk



CHALLENGE 1 OF 9

Design Brief: City for 100 Years

Professional architects start with a brief, not a drawing. Write a brief for a neighbourhood designed to still work brilliantly in 100 years.

***Tip:** A good brief answers: who is it for, what must never go wrong, and what would make it loved, not just used?*

Did you know: Architects use a real idea called "long life, loose fit": buildings designed to be easily changed last far longer than rigid ones.

Rate your own work: Needs work 1 2 3 4 5 Nailed it

One thing I would improve next time:



CHALLENGE 2 OF 9

Section Drawing

Architects do not just draw the outside. Draw a section, a slice through a building as if cut in half, showing the floors, stairs and rooms inside. Include at least 3 floors, how people move between them, and where light gets in.

Did you know: A "section" is one of the three core architect drawings, alongside the plan (from above) and elevation (from the front).

Draw your section here

Rate your own work: Needs work 1 2 3 4 5 Nailed it

One thing I would improve next time:



CHALLENGE 3 OF 9

The Trade Off Triangle

Every project juggles three things: fast, cheap, good. You usually only get two. Pick a building and decide which two you would protect, and what you would sacrifice.

***Tip:** Fast plus cheap means lower quality. Good plus fast means expensive. Good plus cheap means it takes longer.*

Did you know: This is a real rule used across architecture and engineering. Knowing what to sacrifice is a sign of an experienced designer.

Rate your own work: Needs work 1 2 3 4 5 Nailed it

One thing I would improve next time:



CHALLENGE 4 OF 9

Adaptive Reuse

The greenest building is one that already exists. Take an old building (a factory, church or warehouse) and redesign its inside for a brand new use. What was it?

What will it become? What do you keep, and what do you change?

Did you know: London's Tate Modern is a former power station. Adaptive reuse saves the carbon locked inside old buildings.

Sketch your adaptive reuse here

Rate your own work: Needs work 1 2 3 4 5 Nailed it

One thing I would improve next time:



CHALLENGE 5 OF 9

Inclusive Design Audit

Go beyond ramps. Audit a real public place (your school, a station, a shop) and list everything that excludes someone, then redesign three of them.

***Tip:** Barriers hide everywhere: signs only in one language, seating with nowhere to rest a wheelchair, lighting or sound that overwhelms.*

Did you know: 1 in 5 people has a disability, many invisible. Inclusive design assumes difference is normal, not an afterthought.

Rate your own work: Needs work 1 2 3 4 5 Nailed it

One thing I would improve next time:



CHALLENGE 6 OF 9

Site and Climate

Architects design with the weather, not against it. Pick a climate (desert, arctic, tropical) and design a home shaped by it. Which way does it face the sun? How does it stay comfortable without machines? What local materials would you use?

Did you know: Traditional desert homes use thick walls and tiny windows to stay cool. Old design wisdom often beats modern air conditioning.

Design your climate home here

Rate your own work: Needs work 1 2 3 4 5 Nailed it

One thing I would improve next time:



CHALLENGE 7 OF 9

Density Dilemma

A growing town needs 500 new homes. Argue the case: tall flats, or spread out houses? Weigh both, then make your decision and defend it.

Tip: Tall saves land but can feel cramped. Spread gives gardens but eats countryside. What about transport, schools, community?

Did you know: Density is one of the biggest debates in real planning. There is no perfect answer, only trade offs for that place.

Rate your own work: Needs work 1 2 3 4 5 Nailed it

One thing I would improve next time:



CHALLENGE 8 OF 9

Form Follows Function

A famous architect rule: the shape of a building should come from its job. Invent a building for an unusual purpose and let its job decide its shape. A library that floats?

A hospital in space? A school underwater?

Did you know: "Form follows function" was coined in 1896. The Sydney Opera House shells come from the music happening inside.

Draw your building here

Rate your own work: Needs work 1 2 3 4 5 Nailed it

One thing I would improve next time:



CHALLENGE 9 OF 9

Defend Your Vision

Architects face tough panels. Write your pitch, then write the three hardest questions a critic might ask, and answer them. (It is too expensive. It has been done before. It will not last.)

Did you know: The Eiffel Tower was hated by critics who called it ugly. The architect defended it so well it became the symbol of France.

Rate your own work: Needs work 1 2 3 4 5 Nailed it

One thing I would improve next time:
